

DisasterAware: AI Enhanced Concept-Based Fuzzy Reasoning for Scalable Wildfire Decision Support

Abstract—Operational wildfire decision support systems choose from closed catalogues of predefined alternatives, cannot revise their rule bases during an incident, and rarely expose the reasoning behind their recommendations. This article presents DisasterAware, a distributed, concept-driven fuzzy decision support framework built on two separated cores: a physics-based simulation core and a decision core that reads it only through observations. Local agents turn ten confidence-gated features into five decision concepts, a sparse Takagi-Sugeno rule base maps them to six intervention families, and a non-inferential coordination layer ranks regions and shares capacity. The central contribution is the open decision space: a staged adaptation loop tunes rule parameters, instantiates rules on uncovered situations, and, when the base has no answer, prompts a pretrained language model as a schema-constrained proposer. Every modification must clear deterministic verification gates and a satisfying forecast test before it acts, so the doctrine grows from five seed rules only as the incident demands, while a graduated fail-safe protects life-safety orders and every decision stays traceable. The simulation core reproduces the front propagation of four historical fires with 86 to 94 % agreement, and on this ground the full open configuration reduces mean burned area by 82 % against no action, nearly halves the loss of the static rule base, decides in under one % of the cycle, and carries sixteen times more agents at under 1.3 times the cost.

Index Terms—Concept-driven Takagi-Sugeno fuzzy inference, distributed multi-agent reasoning, explainable and uncertainty-aware decision support, open decision space, wildfire decision support systems.

I. INTRODUCTION

Wildfire commanders decide under uncertainty, time pressure, and resource limits, where a poor decision can be irreversible. Decision support systems (DSS) extend what an operator can weigh [1], yet they assume full observability, stationary dynamics, and defined objectives [2], which rarely hold under persistent epistemic uncertainty [3], and AI methods raise accuracy but reason opaquely where critical domains demand traceable outputs [4]. Wildfire makes this concrete: fire frequency, extent, and intensity grow under climate change and the expanding wildland-urban interface [5], impacts outpace conventional response [6], and behavior evolves under weather, terrain, fuel, and suppression feedback [7, 8]. Operational assessments are incomplete and linguistic, which suits fuzzy inference over partial evidence without probability distributions [9].

Many systems cover parts of this task, from optimization based allocation and simulation platforms to geospatial risk assessment and fuzzy reasoning [6, 10]. Mature deployed

systems such as WFDSS [11] and FlamMap [8], and recent hybrid predictive fuzzy models [12], deliver operational value, yet none unites adaptability, explainability, and distributed coordination, and practitioners report using such tools to validate decisions already made rather than to generate options [13, 14].

These limitations consolidate into the three gaps of Table I. The first is documented for closed alternative sets in centralized, single-layer cores [1, 15, 16, 17, 18], where a recent review finds effectiveness evaluation largely absent [19]. The second follows from uncertainty treatments presupposing prior distributions or belief intervals unavailable at cell level in real time [20, 21, 22], from limited use of fuzzy representations in wildfire DSS [9], and from post-hoc attribution that approximates behavior instead of tracing decisions to causes [4, 12]. The third follows from platforms that merge fire modeling with decision logic [11] and from constraint formulations presupposing complete state information [2, 23, 24, 25].

TABLE I
STRUCTURAL GAPS IN WILDFIRE DECISION SUPPORT ARCHITECTURES

Structural Gaps of Existing Wildfire DSS Architectures	
G1	Predefined alternatives in centralized, level-undifferentiated architectures, preventing adaptation to unforeseen conditions and limiting scalability.
G2	Epistemic uncertainty representable only through prior distributions or belief intervals; reasoning exposed only through post-hoc attribution.
G3	Simulation and decision reasoning tightly coupled, constraints managed by offline optimization requiring complete state information; no end-to-end workflow demonstrates real-time, cell-level support.

DisasterAware answers these gaps with a concept-driven fuzzy design whose central contribution is the open decision space: the interventions the system can order grow during the incident, under verification. Three contributions match the gaps of Table I one to one.

Open decision space through adaptive, concept-driven reasoning. To address G1, the fixed menu of alternatives becomes an open decision field: ten observation features aggregate into five decision concepts, so rules are written in operational language. The field is reshaped online on three levels: an evolving Takagi-Sugeno base adapts memberships and consequents [26, 27], term resolution grows where situations are ambiguous, and a generative stage may add intermediate concepts and compose verified physical effects into new interventions; nothing enters the base without a verification gate ending in a reseeded forecast comparison [28]. Local agents do all reasoning, while a non-inferential global layer ranks priorities, assigns resources, and adjusts

acceptance gates.

Sensitivity analysis with traceability and confidence-gated uncertainty. To address G2, explanation and uncertainty handling are structural. Epistemic uncertainty is propagated without prior distributions by blending each concept activation with a persistence-based estimate in proportion to observation confidence. Every decision traces to its concepts, rules, and confidences, and a sensitivity analysis quantifies how the closed-loop parameters affect the outputs, in contrast to post-hoc attribution [4].

End-to-end use-case demonstration. To address G3, the two cores are strictly separated, validated independently, and exercised end to end with an operator dashboard, every intervention screened before fielding. Paired with-DSS and without-DSS runs share initial state, drivers, and random stream, so differences arise from the issued orders alone [13, 14].

II. BACKGROUND

A. Wildfire Dynamics and Spread Modeling

Fire behavior is governed by weather, terrain, and fuel: wind drives the front, humidity and precipitation set fine-fuel moisture, terrain adds slope and aspect, and fuel type and load set combustion intensity [7, 29]. They are summarized by the rate of spread, its direction, and the fire line intensity, which the Rothermel model still expresses in most operational tools[7].

Spread models are physical, empirical, or simulation based, the last propagating a front with an empirical kernel, as in [30] FlamMap [8]. A wildfire DSS needs a testable environment rather than maximal fidelity: reproducible scenarios, observable per-cell state, controllable intervention inputs, and outcomes measurable against not deciding. DisasterAware therefore adopts the hybrid grid model of Section III, which applies Rothermel-type local spread with per-cell state tracking and accepts interventions only at the input level.

B. Decision Support Foundations

A DSS mediates between the input space of what the system can observe and the decision space of what it is authorized to decide, helping an operator choose among actions with uncertain consequences [1]. Wildfire response is the canonical semi-structured case: spread can be modeled and responses scored, yet accountability stays with the commander, and time-critical domains require explicit uncertainty handling [3].

In a closed decision space the alternatives are enumerated in advance and the system can only rank them [15]. Fire routinely produces conditions no predefined alternative covers, and commanders build courses of action from recognized patterns rather than menus [31]. An open space instead constructs new alternatives and verifies each before it acts. The elements exist separately: evolving Takagi-Sugeno models create rules online [27], value-based selectors learn when to change structure [32], and generative models draft

rules outside the current structure [33]. DisasterAware combines the three behind a satisficing acceptance gate [28], so the space stays open while every realized action stays admissible.

In a centralized architecture every observation flows to one reasoning core: communication grows with the domain, latency with communication, and a core failure silences the system [17]. Distributed architectures move reasoning to local nodes, as wildfire DSS reviews recommend [10]. DisasterAware follows this pattern with one restriction: local agents perform all inference and the global layer none, assigning resources and normalizing priorities. The channel carries three compact quantities per region rather than raw fields, keeping coordination auditable and the local path short.

C. The Input Space: Heterogeneous Sensing to Observation Features

Five families of sources feed the framework: meteorological networks and forecasts, spaceborne and airborne imagery with wide coverage but coarse revisit, in-situ sensing with narrow footprints, geographic layers for terrain, fuel, assets, and roads [29], and resource tracking and field reports from the response organization [25]. None observes the fire state directly and they differ in coverage, refresh, and reliability, so the stream is reduced to the ten bounded features of Table II. A quantity is admitted only if observable, bounded, operationally meaningful, and non-redundant; the ten span threat, exposure, feasibility, and timing, so removing one leaves an operational question unsupported.

TABLE II
THE TEN OBSERVATION FEATURES, THEIR SOURCES, AND THEIR RATIONALE

Feature	Source and rationale
Fire intensity	Imagery and in-situ sensing; how intensely a front burns
Spread potential	Fire state and weather; Rothermel rate of spread
Weather severity	Stations and forecasts; fire-weather indices for spread potential
Ignition proximity	GIS road and infrastructure layers; human ignitions cluster nearby
Fuel load	Fuel model layers parameterizing combustible load
Asset exposure	GIS asset and urban interface layers; values-at-risk mapping
Resource accessibility	Terrain, fuel, and access layers; suppression difficulty
Access and road status	GIS road status and field reports; logistics and evacuation egress
Suppression availability	Resource tracking; fireline production rates and available capacity
Temporal urgency	Ignition timing and incident clock

D. Concept-Driven Fuzzy Reasoning

Operational assessments arrive as graded, linguistic judgments, whose formal counterpart is partial membership [9]. A fuzzy set is characterized by a membership function assigning each element a degree of membership, (1). Domain knowledge enters as linguistic rules of the canonical conjunctive form (2), whose activation strength is the conjunction of its antecedent memberships (3).

$$\mu_{\mathcal{A}}: Z \rightarrow [0,1], \mathcal{L} = \{(z, \mu_{\mathcal{A}}(z)): z \in Z\} \quad (1)$$

$$R_r: \text{IF } z_1 \text{ is } \mathcal{A}_{r,1} \text{ AND } \dots \text{ AND } z_n \text{ is } \mathcal{A}_{r,n} \text{ THEN } u \text{ is } \mathcal{B}_r \quad (2)$$

$$\alpha_r = \min_i \mu_{\mathcal{A}_{r,i}}(z_i) \quad (3)$$

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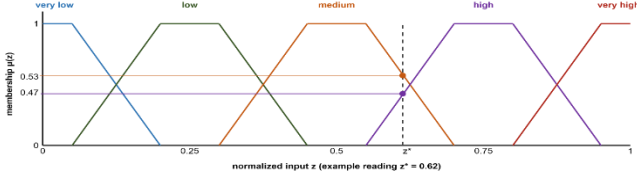


Fig. 2. Five-term trapezoidal membership partition with an example reading.

Takagi-Sugeno inference turns rule activations into a crisp action, replacing the consequent set with a function of the inputs and taking the output as a firing-strength weighted average [26]. DisasterAware adopts the zero-order Takagi-Sugeno form, every consequent a constant intensity on the unit interval, stated in full in Section IV. The choice is architectural: candidates are judged inside shadow forecasts run thousands of times per incident, so centroid integration per evaluation would be too costly, the outputs are intensity commands, and adaptation needs numeric consequents with predictable effect while antecedents stay linguistic. Fig. 1 depicts 5-term trapezoidal partition; adjacent terms overlap so every reading activates at most two terms.

Applying fuzzy inference directly over a rich observation vector is combinatorially infeasible: a complete rule base over the full feature set has the size given in (4): ten features with five terms each demand nearly ten million rules, which no expert can author or audit. A conceptual middle layer reduces this, six base concepts summarizing physically related features and five decision concepts answering, each exactly once, the operational questions commanders ask [6, 31]; a single top concept, operational priority, serves cross-region coordination. Rules are written over the five decision concepts only, so the rule burden collapses from a product to a sum (5). The antecedent space shrinks to 3125 combinations, only the active subset instantiated, and the concepts double as the traceability vocabulary: every recommendation reads back over fire threat level, suppression feasibility, asset exposure risk, intervention urgency, and evacuation pressure.

$$|R| = N_{\text{term}}^{N_{\text{feature}}} \quad (4)$$

$$N_{\text{term}}^{N_{\text{feature}}} \rightarrow N_{\text{concept}} \cdot N_{\text{term}}^{N_{\text{input}}} + N_{\text{term}}^{N_{\text{concept}}} \quad (5)$$

E. The Decision Space: Interventions and Their Evaluation

The rules assign intensities to concrete actions. The decision space is built from six intervention families, each applied with an intensity over the grid, so every rule binds to an executable order [6]: suppression effort [25], resource deployment [23], containment line, asset protection [16], evacuation, and public warning [18]. The list is closed, keeping the vocabulary interpretable; openness enters through new rules and compositions over the same types. Beneath them the simulator verifies physical effects (wet, clear, ignite, coat, draft, evacuate, prime) and three channels built on them, water drafting, retardant drop, and tactical burn, validated although no seed rule orders them. Inaction is the zero-intensity point of every family, so the cost prices restraint as it prices action.

A candidate intervention is evaluated through its expected cost, following the cost-plus-loss principle of wildfire economics [6, 25]. Global minimization over a spatial decision field is intractable under real-time constraints and needs

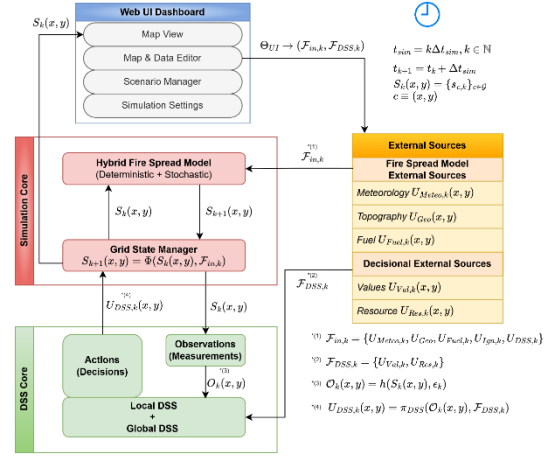


Fig. 1. DisasterAware system block diagram (simulation focused).

information incomplete observation cannot supply, so acceptance is satisfying [28]. The dominant uncertainty is epistemic, the state under-observed rather than noisy [20], and Bayesian or Dempster-Shafer treatments presuppose distributional knowledge that real-time sensing lacks [21]. Explanation is structural, over explicit concepts and rules, not post-hoc attribution [4]. Generative proposers may be wrong or unsafe, so generated artifacts are untrusted by construction and admitted only after verification; Section IV develops this machinery.

III. DISASTERAWARE FRAMEWORK AND SIMULATION CORE

A. Overall Architecture

Fig. 2 presents the DisasterAware architecture: external data sources, the simulation core, the DSS core, and the operator dashboard communicate through structured interfaces. The simulation core is authoritative, since cells carry burning status, fuel mass, an intensity proxy, and time since ignition and no external module may overwrite this state; the DSS core reasons over derived observables, and the dashboard admits only input-level operator actions such as pause, rollback, ignition injection, and resource changes.

B. External Inputs and Decision Context

The external input set (6) collects the parameters that influence wildfire evolution at step k : meteorological forcing (temperature, humidity, wind, gusts, precipitation), topography (elevation, slope, aspect, accessibility), fuel conditions (class, load, moisture), ignition injection, and the DSS intervention parameters (capacity, availability, effectiveness, travel time). A separate contextual set guides decision generation only: a value layer aggregating buildings, critical facilities, population, and evacuation access into a protection priority score, and a resource layer of available capabilities. It never touches fire dynamics, keeping modeling and reasoning separate.

$$\mathcal{F}_{in,k} = \{U_{Meteo,k}, U_{Geo}, U_{Fuel,k}, U_{Ign,k}, U_{DSS,k}\} \quad (6)$$

C. State Representation and Transition

The landscape is discretized into a rectangular grid, and at

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each simulation step the wildfire state is the collection of local cell states (7), each cell carrying burning status, fuel mass, an intensity proxy, and time since ignition; environment and interventions enter only as external inputs. The global evolution (8) is a structured transition operator over burning status, fuel mass, and intensity, illustrated in Fig. 3. The operator itself is deterministic; stochastic variability enters only through exogenous input fields such as wind gusts, and, in one controlled channel, through seeded ember spotting.

$$S_k = \{s_k(x, y) | (x, y) \in \mathcal{G}\}; s_k = \begin{bmatrix} B_k(x, y) \\ F_{load,k}(x, y) \\ I_k(x, y) \\ \tau_k(x, y) \end{bmatrix} \quad (7)$$

$$S_{k+1}(x, y) = \Phi(S_k(x, y), \mathcal{F}_{in,k}) \quad (8)$$

D. Hybrid Fire Spread Model

A cell burns through persistence while fuel remains, through propagation from burning neighbors, or through external ignition. Propagation fires when the influence accumulated over the 8-connected neighborhood exceeds a calibrated threshold, each neighbor contributing its Rothermel-type rate of spread from wind, slope, aspect, moisture, and fuel type [7], weighted toward the wind direction. Ember spotting, the only stochastic element and driven by the seeded scenario stream, carries fire several cells downwind under strong dry wind and across narrow barriers such as rivers, roads, and containment lines. Precipitation raises dead fuel moisture above the extinction level so a front dies out gradually, and the equilibrium moisture mode lets diurnal weather modulate flammability.

E. Fuel Mass, Suppression, and Intensity

Fuel mass decreases through combustion and suppression, never below zero. Suppression converts capability into a normalized fuel reduction, gated by capacity and availability, decaying with travel time and poor access, and attenuated by local fire intensity; it never overwrites the burning state but acts on the fuel, which shapes persistence and propagation. A local intensity proxy from fuel, wind, and slope modulates suppression effectiveness and feeds the decision layer without controlling ignition or spread.

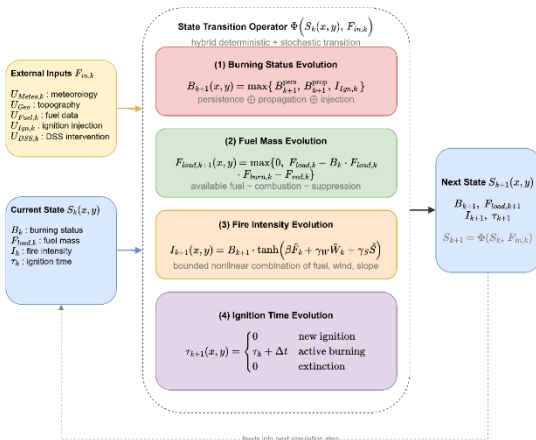


Fig. 3. State transition block diagram.

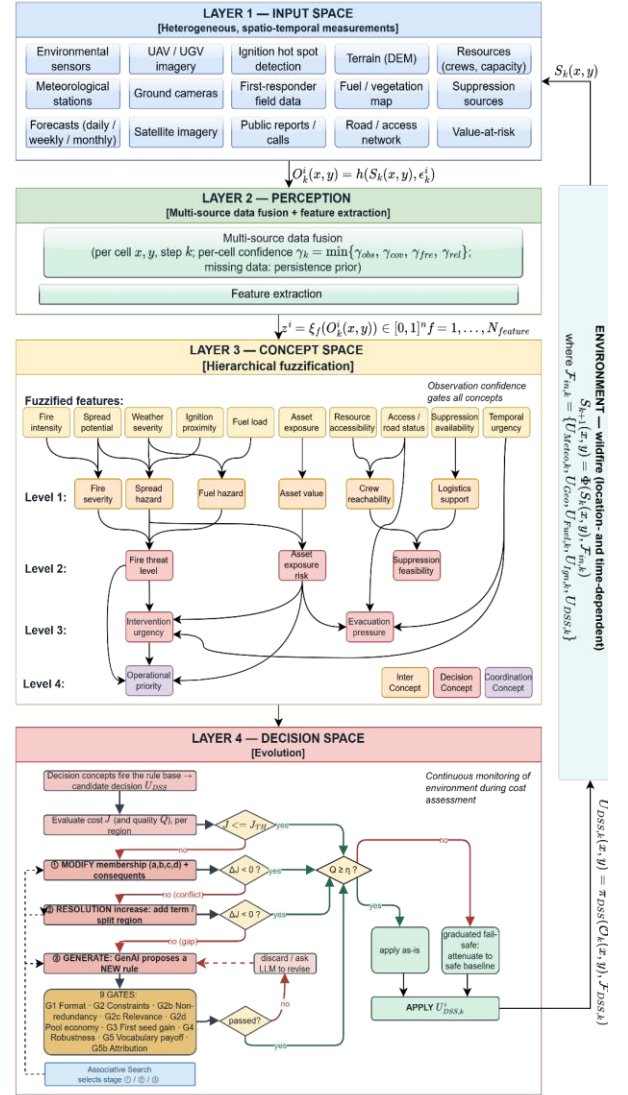


Fig. 4. DisasterAware system block diagram (DSS focused).

F. Interaction, Counterfactual Replay, and the DSS Interface

The operator acts only through bounded input modifications, never through the state, so interaction stays prospective and reversible. Per-step snapshots, including the random number generator, enable exact counterfactual replay: runs from the same snapshot differ only in the orders under test, the basis of the with and without DSS comparisons of Section V. The DSS closes the loop through the observation interface (9): the observation function projects the state onto observable information, injecting epistemic uncertainty at this point only, and the aggregated intervention field re-enters the simulation only as one component of the external input set.

$$\begin{aligned} S_k(x, y) &\rightarrow O_k(x, y), \mathcal{F}_{DSS,k} \rightarrow U_{DSS,k}(x, y) \rightarrow \Phi(\cdot) \\ &\rightarrow S_{k+1}(x, y) U_{DSS,k}(x, y) \\ &= \pi_{DSS}^i(O_k(x, y), \mathcal{F}_{DSS,k}) \subset \mathcal{F}_{in,k} \end{aligned} \quad (9)$$

IV. DECISION SUPPORT SYSTEM

A. Four-Layer Pipeline

Fig. 4 presents a Local DSS agent as a four-layer flow: the

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input space collects the measurements of Section II-C, the perception layer fuses them into bounded features with explicit confidence, the concept space abstracts them, and the decision space generates, adapts, and evaluates interventions. N agents run this loop over their own regions, coordinated by the Global DSS through priorities and shared resource pool.

B. Observation, Features, and Confidence

Each Local DSS agent receives at step k the region-restricted observation (10), where the observation function projects the authoritative state onto the cells of the region and injects the epistemic uncertainty. This uncertainty stems from missing knowledge rather than randomness (an occluded camera, a satellite revisit gap, an offline sensor), and no distribution over the error is assumed anywhere. Every observed cell carries a confidence composed conservatively from four factors, observability, coverage, freshness, and reliability, as the minimum (11) rather than an average. Confidence bounds the disturbance instead of describing it statistically: the observation error magnitude is limited by the complement of the confidence (12), so full confidence is exact, vanishing confidence admits the worst case, and intermediate values degrade gracefully. The agent then computes the ten features (13), each inheriting a feature-level confidence from the observation fields its extraction consumes, so the pair of feature value and feature confidence is the complete interface between perception and reasoning.

$$O_k^i(x, y) = h^i(S_k(x, y), \varepsilon_k^i(x, y)), \quad (x, y) \in \mathcal{G}^i \quad (10)$$

$$\gamma_k^i(x, y) = \min \{\gamma_{obs}^i, \gamma_{cov}^i, \gamma_{fre}^i, \gamma_{rel}^i\} \quad (11)$$

$$|\varepsilon_k^i(x, y)| \leq (1 - \gamma_k^i(x, y)) \cdot \varepsilon_{max} \quad (12)$$

$$z_{f,k}^i = \xi_f(O_k^i) \in [0, 1], \quad f = 1, \dots, N_{feature} \quad (13)$$

C. Concept Space and Confidence Gating

The conceptualization layer of Fig. 5 maps physically related features into six base concepts, the base concepts into three mid-level and two higher-level decision concepts, and finally into the single coordination concept. The activation of a base concept is the weighted sum of the term memberships of its features, and the activation of every higher concept is the weighted sum of the activations of the concepts in its set, with

nonnegative weights normalized to one (14). Raw activations are never consumed directly. Every activation is gated by its observation confidence and blended with a persistence-carried prior (15): a reliable observation passes unchanged, an uninformative one falls back to the decayed previous estimate, and an unobserved concept fades geometrically. The trust-caution transition is linear with no threshold to tune, and the activation snaps back as soon as confidence returns.

$$a_{n,k}^{(1),i} = \sum_{f \in Z_n} \omega_{n,f} \cdot \mu_{\mathcal{A}_{n,f}}(z_{f,k}^i) \quad (14)$$

$$a_{n,k}^{(l),i} = \sum_{m \in \mathcal{C}_n^{(l)}} \omega_{n,m}^{(l)} \cdot a_{m,k}^{(l-1),i}, \quad l = 2, 3, 4$$

$$a_{n,k}^{eff,i} = \gamma_{n,k}^i \cdot a_{n,k}^{obs,i} + (1 - \gamma_{n,k}^i) \cdot \rho \cdot a_{n,k-1}^{eff,i}, \quad (15)$$

$$\gamma_{n,k}^i = \min \{\gamma_{f,k}^i : f \in Z_n\}$$

D. Rule Base and Candidate Intervention

Decision rules are written exclusively over the five gated decision concepts in the canonical conjunctive form (16): each antecedent clause tests one concept against one linguistic term and the consequent assigns a singleton intensity to each family. A rule fires with the strength of its weakest satisfied clause, and the candidate is assembled family by family from the rules that fire (17), with the denominator floored at one so a weakly supported conclusion yields a weak order. The base is sparse: of 3125 combinations it holds only the seeded doctrine and what adaptation adds, at no cost to coverage, since adjacent terms overlap and the aggregation interpolates. The seeded base satisfies the coverage condition (18), so some rule fires above the coverage floor of 0.05 at every reachable reading; below the floor the layer falls back to the watchful posture and adaptation is triggered. Each concept has a primary family, suppression feasibility acting as gatekeeper, and a run starts from five seed rules.

$$R_r: \text{IF } a_{1,k}^{eff,i} \text{ is } \mathcal{A}_{r,1} \text{ AND } \dots \text{ AND } a_{5,k}^{eff,i} \text{ is } \mathcal{A}_{r,5} \text{ THEN } U_L^i \quad (16)$$

$$u_{m,k}^i = \frac{\sum_r \alpha_r v_{r,m}}{\max(1, \sum_r \alpha_r)} \quad (17)$$

$$\alpha_r = \min_{n \in \text{ant}(r)} \mu_{\mathcal{A}_{r,n}}(a_{n,k}^{eff,i})$$

$$\min_{a^{eff,i} \in [0,1]^5} \max_r \alpha_r(a^{eff,i}) \geq \alpha_{min} > 0 \quad (18)$$

E. Cost, Quality, and Satisficing Acceptance

The decision and adaptation flow of Layer 4 is summarized in Fig. 6. Every candidate is judged on a shadow copy of the simulation before anything reaches the field: the cost asks what it is expected to cost the incident, the quality whether it serves the concepts that demanded it.

The cost is the weighted five-term sum (19) over burned area, asset loss, population exposure, response effort, and response delay, each normalized by a scenario reference so the five become summable: burned area through a rational form that keeps every saved cell worth saving, population as person-steps inside burning cells, response as the time integral of committed force, and delay against a sixty-minute reference. The candidate is held constant over a forecast horizon and compared with a no-action shadow run from the same state, and acceptance is satisficing (20): the bound is

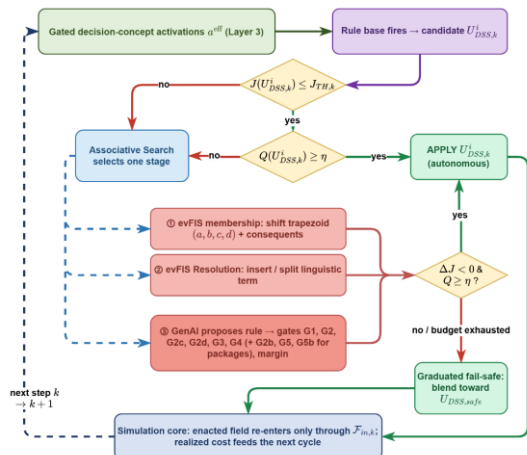


Fig. 6. Layer 4 decision and adaptation flow.

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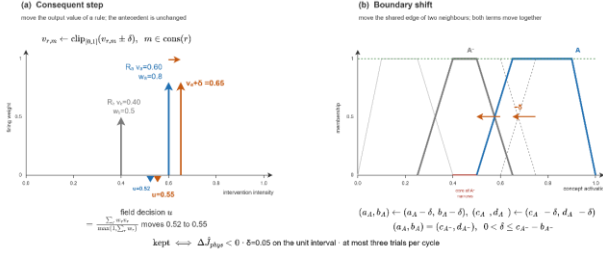


Fig. 8. Consequent step and boundary shift.

the tighter of a fixed ceiling of 0.35 and a required relative gain over the no-action forecast, so cheap inaction demands a stronger candidate; failing the test leaves the standing orders in force while adaptation runs. Cross-run comparisons read the physical cost only, because a run that never intervenes carries no response cost and a total-cost comparison would reward inaction.

$$J = \frac{\sum_j w_j J_j}{\sum_j w_j}, j \in \{burn, asset, pop, resp, delay\}$$

$$J_{burn} = \frac{\frac{N_{burned}}{\rho_{burn} \cdot N_{burnable}}}{1 + \frac{N_{burned}}{\rho_{burn} \cdot N_{burnable}}}$$

$$J_{asset} = \frac{V_{lost}}{V_{total}} J_{pop} = \min\left(1, \frac{P_{steps}}{P_{total} \cdot N_{decisions}}\right) \quad (19)$$

$$J_{resp} = \min\left(1, \frac{C_{steps}}{C_{ref} \cdot N_{decisions}}\right)$$

$$J_{delay} = \min\left(1, \frac{\tau}{\tau_{reference}}\right)$$

$$\hat{J}_k(U_{DSS,k}) \leq \min(J_{TH}, (1 - \delta_{gain}) \cdot \hat{J}_k(0)) \quad (20)$$

The quality measures what the cost cannot. For each scored concept, the gated activation is compared with the strongest intensity the candidate assigns to the family that answers it, and the quality is one minus the mean mismatch (21). Over-suppression at low threat is penalized as much as ignoring high evacuation pressure, so misdirection is caught before its consequences appear; suppression feasibility, which states whether action is possible rather than how much is required, modulates rather than demands. Below the quality gate of 0.60 the graduated fail-safe attenuates the candidate toward a pre-authorized conservative posture (22) in proportion to the deficit: the offensive families, suppression effort, resource deployment, containment line, and asset protection, are scaled down, while evacuation and public warning are never reduced. The transition is continuous and reversible as observation quality recovers, and if the adapted candidate is still forecast worse than doing nothing on the physical cost, a no-harm rule withholds the offensive allocation over a lookahead, so interventions maturing in thirty to sixty minutes are not punished prematurely.

$$Q_k = 1 - \frac{1}{4} \sum_{n \in C_Q} |a_n^{eff} - \max_{f \in F(n)} u_f|, \text{ accept} \Leftrightarrow Q_k \geq \eta \quad (21)$$

$$U_{DSS,k}' = \lambda_k \cdot U_{DSS,k} + (1 - \lambda_k) \cdot U_{DSS,safe}, \lambda_k \in [0,1] \quad (22)$$

F. Open Decision Space: Staged Adaptation

Adaptation is organized as three stages of increasing

invasiveness, engaged only when the standing decision demonstrably falls short: a cost deficit, a coverage gap while fire is alive, or renewed spread under applied orders. Stage 1, the evolving-fuzzy parameter layer, tunes rules already in the base with two bounded moves illustrated in **Fig. 7**: the consequent step (23) nudges the intensity of a rule by a fixed 0.05 within the unit interval, and the boundary shift moves the shared edge of two neighboring terms while preserving the partition of unity. Every trial is kept only when the forecast physical cost strictly falls (24) and rolled back otherwise; trials act on a working copy, so the seed base is never modified. Stage 2 grows the base: it writes a rule on the uncovered combination of the two most active concepts, and where a covered combination is ambiguous it inserts a narrow term at the crisp reading (3125 to 3750 combinations at first insertion); inserted terms stay outside the Layer-3 aggregation.

$$v_{r,m} \leftarrow \text{clip}_{0,1}(v_{r,m} \pm \delta), \quad m \in \text{cons}(r) \quad (23)$$

$$\text{kept} \Leftrightarrow \Delta \hat{f}_{phys} < 0 \quad (24)$$

Stage 3, the generative layer, adds knowledge that neither tuning nor instantiation can produce. The proposer is a pretrained language model used in context only, with no task-specific training: it receives a deterministic situation summary, the active rule base in IF-THEN form, and the coverage verdict, and returns exactly one proposal in the canonical rule form. The stage produces four kinds of output: a plain rule, a macro intervention blending existing interventions, an intermediate concept that joins the hierarchy like a native one, and a clause actuator carrying its own map geometry. Its vocabulary is deliberately primitive, the verified action list, a geometry of sectors and distance bands, and the map affordances, so an intervention is composed rather than picked from a menu: given a lake near the front, the stage can compose an actuator that drafts from it and wets the fuel ahead of the head, which no seed rule contains; the basis itself is not proposable, so the space stays closed at effects yet open at interventions. Every proposal is untrusted by construction and passes the applicable gates of **Fig. 9**: G1 format, G2 vocabulary, ranges, and duplication, G2b concept non-redundancy, G2c relevance, and G2d pool economy, all without a simulation step; G3 and G4 then demand that the forecast physical cost strictly fall on two independently seeded rollouts (25), G5 requires a package to clear an additional

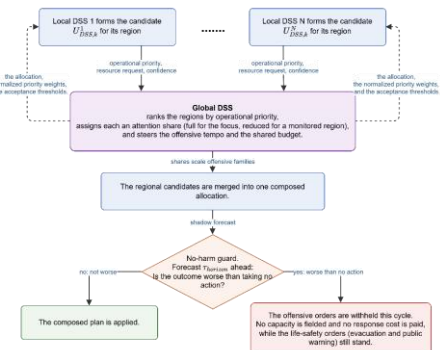


Fig. 7. Global coordination and the no-harm fail-safe.

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$$\Delta f_{phys}^{(v)} = \hat{f}_{phys}^{(v)}(\mathcal{R} \cup \{r'\}) - \hat{f}_{phys}^{(v)}(\mathcal{R}), \quad v \in \{V_a, V_b\} \quad (25)$$

$G3: \Delta f_{phys}^{(V_a)} < 0; G4: \Delta f_{phys}^{(V_b)} < 0$

G. Global Coordination and the No-Harm Guard

The Global DSS performs no inference of its own: each Local DSS sends up its operational priority, its resource request, and a confidence summary, and the coordinator returns the allocation, the normalized priority weights, and the acceptance thresholds in force. Regions are ranked by operational priority, and each receives an attention share (26): a region near the top priority keeps the full share, a monitored region gets a reduced share floored at 0.50 and a stricter quality threshold for offensive orders; evacuation and warning are untouched. The merged plan is forecast by the no-harm guard against taking no action, and offensive orders are withheld when it is clearly worse while life-safety orders stand. Fig. 8 summarizes the coordination and the guard. Every decision remains traceable end to end, with fired concepts, rules, strengths, input confidences, and gate verdicts all recorded, so explanation is structural rather than post-hoc.

$$\sigma_i = \begin{cases} 1, & \text{if } p_i \geq \tau_{att} p_{max} \\ 0.5 + 0.5 \frac{p_i}{p_{max}}, & \text{otherwise} \end{cases} \quad (26)$$

V. RESULTS

A. Evaluation Criteria

The technical tradition evaluates the predictive models inside a tool on real fires, while the decision-support tradition evaluates its effect on the decision itself, and most wildfire DSS evaluations measure adoption rather than decision improvement [19]. Both are addressed through the eight criteria of Table III, fixed in advance and audited at the end of this section.

TABLE III
EVALUATION CRITERIA FOR WILDFIRE DECISION SUPPORT SYSTEMS

Criterion	Measurement in the literature
Predictive fidelity	Retrospective hindcasts against historical fires; spatial-overlap and arrival-time scores
Decision quality	Outcome deltas against no-action and baseline policies on matched scenarios
Timeliness	Decision-cycle wall time; detection-to-first-order time

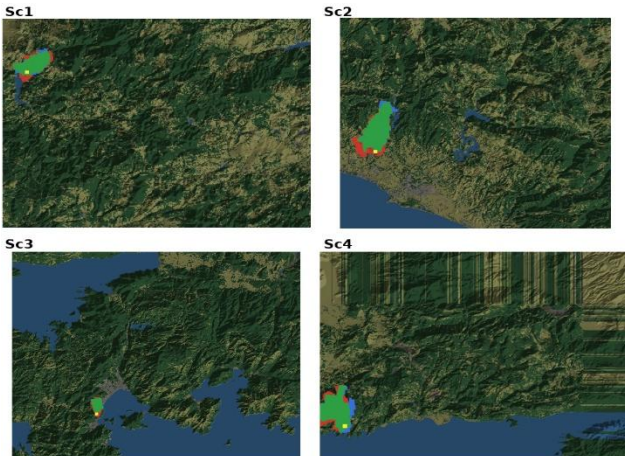


Fig. 10. Hindcast agreement maps (green: predicted burn; red: simulated only; blue: observed only; yellow: ignition).

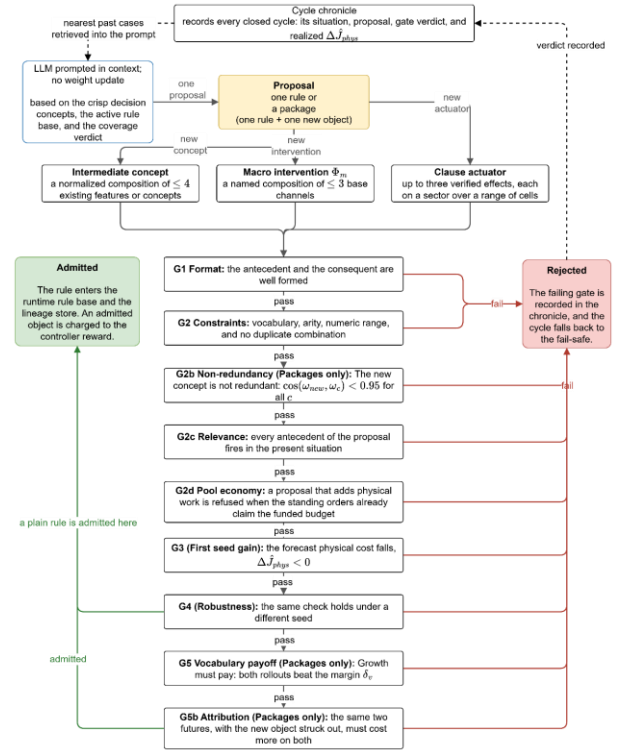


Fig. 9. Generative proposal and the verification gates.

Robustness	Performance versus outage and noise level; fail-safe engagement statistics
Uncertainty handling	Ensemble and input-uncertainty propagation studies
Adaptability	Learning curves; coverage growth; rediscovery of withheld knowledge
Explainability and auditability	Trace completeness; interpretability as an adoption gate
Scalability	Runtime and quality versus grid size and agent count

B. Validation of the Simulation Core

The simulation is tested by blind hindcast: it receives only the real inputs of a documented fire, with no tuning on that fire, and its output is compared with the satellite record. Terrain comes from the Copernicus GLO-30 model, fuel from ESA WorldCover 2021, weather from ERA5 reanalysis, and ignition points and reference burn from NASA FIRMS VIIRS detections. Four documented fires in Türkiye cover different fuel, terrain, and wind conditions: Sc1 Manavgat and Sc2 Marmaris (July 2021, 48 h), Sc3 Milas-Kemerköy (August 2021, 72 h), and Sc4 Çanakkale-Kayadere (August 2023, 48 h). Because the core models free spread without suppression, a free run overpredicts the final area, so three suppression-robust criteria are used: the probability of detection (27), accepted at 0.70 for calibrated growth models [34]; the front position error relative to the equivalent burn radius (28), (29); and the arrival-time agreement (30), the direct rate-of-spread test. Table IV reports the metrics over 50 replicates and Fig. 10. Hindcast agreement maps (green: predicted burn; red: simulated only; blue: observed only; yellow: ignition). the agreement maps: overall agreement is 86 to 94 % and detection probability 85 to 97 %, the reliability of a well-behaved

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Fig. 11. Physical cost over ignitions and resource level, as a share of the free burn.10

$$POD = \frac{|FR_{Sim} \cap FR_{Re}|}{|FR_{Re}|} \quad (27)$$

$$D(x, \mathcal{G}_{Set}) = \min_{y \in \mathcal{G}_{Set}} \|x - y\| d_{Front} \quad (28)$$

$$= \frac{\sum_{x \in \partial FR_{Sim}} D(x, \partial FR_{Re}) + \sum_{y \in \partial FR_{Re}} D(y, \partial FR_{Sim})}{|\partial FR_{Sim}| + |\partial FR_{Re}|}$$

$$R = \sqrt{(|N_{burned}| A_{cell})/\pi}, TH_{Front} = 1 - \frac{d_{Front}}{R} \quad (29)$$

$$\tau_{MAE} = \frac{1}{N_{cells}} \sum_{c \in FR_{Sim} \cap FR_{Re}} |t_c^{Sim} - t_c^{Re}|; N_{cells} \quad (30)$$

$$= |FR_{Sim} \cap FR_{Re}| TH_{MAE}$$

$$= 1 - \tau_{MAE}/T$$

TABLE IV
HINDCAST AGREEMENT METRICS PER HISTORICAL CASE

Metric	Sc1	Sc2	Sc3	Sc4
Front distance (m)	329	305	113	391
Arrival-time MAE (h)	1.46	1.39	4.04	2.58
Front agreement	82%	90%	89%	89%
Time agreement	90%	98%	71%	96%
Probability of detection	92%	93%	97%	85%
Overall success	88%	94%	86%	90%

C. Experimental Setup and Sensitivity Analysis

Synthetic scenarios use 200 by 200 grids at 90 m resolution with terrain, settlements, roads, and capacity-limited bases; historical scenarios reconstruct the documented fires. Unless stated otherwise the engine runs with the thresholds of Section IV, a twelve-minute decision cycle, a shadow forecast basis of at least 45 minutes, and 50 Monte Carlo runs per condition, sensitivity being assessed one parameter at a time over five independent worlds, 880 runs in total. **Fig. 11. Physical cost over ignitions and resource level, as a share of the free burn.** maps the physical cost as a share of the free burn: at a tenth of the pool the fire is lost throughout (77 to 83 %) and with a full pool and few ignitions it is held (14 to 32 %), so the sweeps run at the decidable midpoint, four simultaneous ignitions with a quarter of the pool. Observation decides light fire loads and resources heavy ones, neither substituting for the other, and the satisficing bound sets how often the ladder engages, 42 % of cycles at 0.35.

D. Performance of the Decision Layer

Four configurations face identical fires: T0 issues no orders and fixes the cost of inaction; TF5 runs the decision layer on the minimal seed base of five rules; TF5+Ev adds the evolving-fuzzy stages 1 and 2; and the full configuration runs

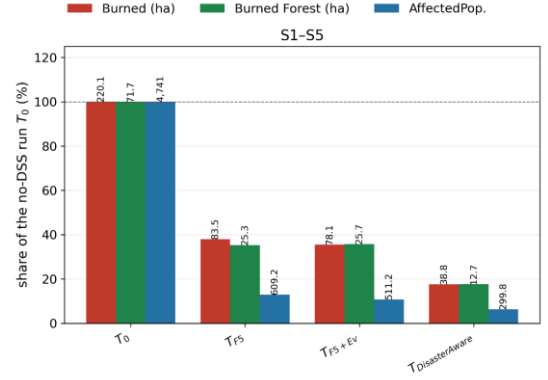


Fig. 11. Average burned-area, burned-forest, and affected-population evolution of the four configurations under the five scenarios.

the complete staged adaptation, the open decision space in full. Every seed generates one world replayed identically by all configurations, so differences come from the decision mechanism alone; evaluation seeds are disjoint from development. Comparisons are read at a common six-hour checkpoint on two costs (31): the physical cost against the no-intervention reference, and the total cost including response and delay. Five scenario families span remote forest and the wildland-urban interface under sufficient and limited resources, the fifth repeating the full stress scenario under degraded observation, all at critical fire weather. Table V reports the ladder, and **Fig. 12** the average evolution. TF5 lowers the mean burned area from 220 ha under free burn to 84 ha (62 %) and the mean total cost from 0.446 to 0.185; TF5+Ev adds a consistent gain to 78 ha and 0.169, the saving coming from the delay term, since tuned consequents commit the same resources sooner. The full configuration halves the burned area again to 39 ha, an 82 % reduction, lowers the affected population from about 511 to 300, and reaches a total cost of 0.112, lowest in every scenario family, while the intermediate rung improves the mean but not every scenario. Under degraded observation the full configuration loses ten % and remains the best of the four, the confidence gates derating offensive orders instead of cancelling them.

$$J_{phy} = \frac{\sum_j w_j J_j}{\sum_j w_j}, \quad j \in \{burn, asset, pop\} \quad (31)$$

$$J_{Tot} = \frac{\sum_j w_j J_j}{\sum_j w_j}, \quad j \in \{burn, asset, pop, resp, delay\}$$

TABLE V
BURNED AREA PER SCENARIO AND MEAN DECISION COSTS AT THE SIX-HOUR CHECKPOINT

Scenario	T0	TF5	TF5+Ev	Full
S1 burned (ha)	243.3	96.7	84.7	22.8
S2 burned (ha)	242.2	157.3	154.6	84.8
S3 burned (ha)	178.9	28.0	30.5	11.4
S4 burned (ha)	218.1	66.2	54.6	35.7
S5 burned (ha)	218.1	69.3	66.3	39.3
Mean Jphys	0.506	0.163	0.151	0.090
Mean Jtotal	0.446	0.185	0.169	0.112

E. Open Decision Space in Operation

Adaptation runs only on demand. Over 160 runs and 7130 decision cycles, stage 1 kept 129 of 616 consequent-step trials

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(21 %), stage 2 kept 40 of 359 (11 %) on antecedent cells left empty while fire was alive, and stage 3 was invoked 232 times with 47 admissions, 36 plain rules and 11 vocabulary packages, 42 rules surviving pruning. Of 157247 candidate interventions 151745 reached the field: the fail-safe engaged in 5338 region cycles and the no-harm veto held the offensive channels in 113. **Fig. 13** shows the two evolving-fuzzy funnels and the generative cascade. Low acceptance rates are intended: trials are cheap and rejected safely on the forecast, and the selectivity of the gates is what makes the admitted edits trustworthy. Table VI lists the generative products: seven macro interventions and one intermediate concept, each born at a stated moment of a specific incident under a logged cost deficit. The three base channels they draw on, water drafting, aerial retardant drop, and tactical burn, are validated by the physics although no seed rule orders them, so a macro composes a capability the seed doctrine ignored rather than a new physical effect. The names describe intent rather than composition, and one product is listed under a corrected name, downwind_containment_shield, since its logged definition carries no igniting channel. The generated concept interface_exposure names the pressure building where exposed assets and evacuation demand meet, joins the hierarchy like a native concept, and is consumed by an admitted rule protecting the wildland-urban interface, so a concept the system introduced and a rule that uses it enter through the same gates.

TABLE VI
THE GENERATIVE PRODUCT LEDGER: EVERY OBJECT CREATED, WITH ITS DEFINITION

Product	Definition
Action-list channels (3)	tactical burn (counter-fire strip ahead of the front); water drafting (capacity near water); retardant drop (coats fuel ahead)
Macros (7)	downwind_containment_shield: containment 1.0 + suppression 0.7; downwind_retardant_shield: retardant 1.0 + containment 0.8; drafting_sustained_attack: drafting 1.0 + suppression 0.9; head_knockdown: drafting 1.0 + suppression 0.9 + retardant 0.6; wet_containment_line: drafting 1.0 + containment 0.8; counterfire_strip: tactical burn 1.0 + containment 0.6; drafting_retardant_shuttle: retardant 1.0 + drafting 0.6
Concept (1)	interface_exposure = 0.5 asset_exposure_risk + 0.5 evacuation_pressure; consumed by admitted rule G14 (IF interface_exposure VH AND suppression_feasibility VH THEN asset_protection 0.9, containment_line 0.8, resource_deployment 0.7)

A second campaign tests whether the stage can create

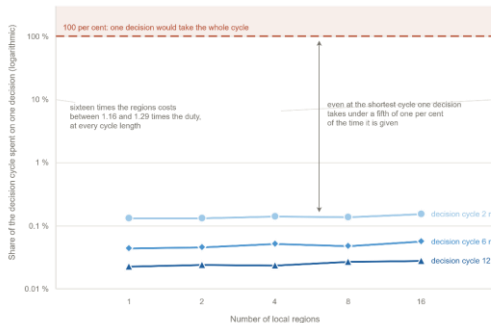


Fig. 14. Decision-layer cost against the number of local agents, as a share of the cycle.

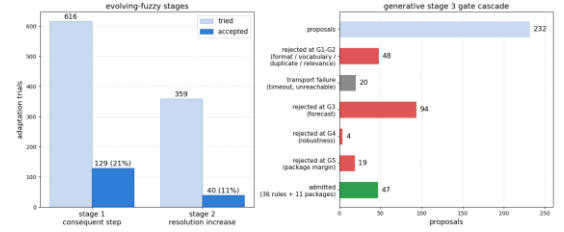


Fig. 13. Left: evFIS trials and acceptances at stages 1 and 2. Right: the generative gates over proposals.

interventions carrying their own geometry, orders stating where on the map to act and how, in terms no base family expresses. Five seeds ran in two arms over 205 decision cycles under the same gates; the stage was entered 35 times and admitted 19 proposals, among them 7 of 14 clause actuators, 5 further intermediate concepts, and 1 further macro. The actuators form two families: first-cycle actuators coat assets and populated cells before the fire arrives, a preventive tactic absent from the seed doctrine, and minute-48 actuators draft from the water body and evacuate behind the front, raising the yield of committed crews on a spent pool. One was later ordered by a separately admitted rule and another is invoked through a concept the same campaign created.

F. Timeliness and Scalability

One decision covers the whole loop, including the two shadow forecasts of the acceptance test and any adaptation stage that runs, and takes a median 274 ms against a twelve-minute cycle, 909 ms at the ninetieth %ile and 1.28 s at the slowest, under two thousandths of the time available; the first offensive order is fielded at the first decision, two minutes after ignition. **Fig. 14. Decision-layer cost against the number of local agents, as a share of the cycle.** reports the decision-layer cost for one to sixteen regions at cycles of two, six, and twelve minutes: every point lies between 0.02 and 0.16 % of the cycle, and sixteen times the agents cost only 1.16 to 1.29 times the share, since each agent reasons over its own region. Six times the forecast horizon raised the median by 1.4 and four times the grid by 6.9, so scalability is met on the axis the architecture is built for while domain size remains the binding limit.

G. Compliance with the Evaluation Criteria

Predictive fidelity rests on four blind hindcasts on real inputs, with front agreement of 86 to 94 % and detection probability of 85 to 97 %. Decision quality follows from burned area falling from 220 to 39 ha (82 %) and physical cost from 0.506 to 0.090 against no action, the full configuration lowest in every scenario. Timeliness is met with a median decision of 274 ms against a twelve-minute cycle, scalability with one decision costing 0.02 to 0.16 % of the cycle and sixteen times the agents costing at most 1.29 times, and robustness with the full configuration losing only 10 % under degraded observation while remaining the best tested. Uncertainty handling is structural, every feature carrying a confidence that gates its concept; adaptability shows in 129 of 616 stage-1 and 40 of 359 stage-2 trials kept and 47 generative

admissions plus seven clause actuators; explainability in every edit, rejection, and generated object being logged with its trigger, gate, and consuming rule. Four criteria are met in full and four on the axis measured here: the hindcast base covers four fires, no external baseline is compared, noise and communication failure are untested, operator-facing interpretability is not evaluated, and uncertainty is carried into the decision but not propagated as an ensemble.

VI. CONCLUSION AND FUTURE WORK

This article presented DisasterAware, a distributed wildfire decision support framework in which concept-driven fuzzy reasoning operates over two separated cores. Ten confidence-gated features feed five decision concepts, a sparse Takagi-Sugeno rule base sets intervention intensities, satisficing acceptance and a graduated fail-safe screen every order, and a non-inferential coordinator ranks the local agents. When the standing decision falls short, a three-stage ladder adapts the parameters, the resolution, and finally the vocabulary of the base, every change passing the same gates. Against criteria fixed in advance, the open decision space starts from five seeded rules and grows only as the field demands, reaching 40 rules, seven macros, one concept, and, in a separate campaign, seven clause actuators, every addition passing a forecast test. It halves the burned area of the static base, sixteen times the agents cost at most 1.29 times per decision, every fielded order traces to its concepts, rules, and confidences, and under degraded observation the framework loses ten percent while remaining the best tested. On an independently validated core, whose four blind hindcasts agree with the satellite record at 86 to 94 percent, the decision core lowers burned area from 220 to 39 ha and physical cost from 0.506 to 0.090 at 274 ms per decision. Three boundaries apply: comparisons are internal, since no deployed tool generates interventions over an open decision space; the hindcast base covers four fires; and generative proposals are not bitwise reproducible, although model, prompt, and seeds are pinned and deterministic gates judge every proposal with a logged verdict. The evaluation is entirely simulated, and four studies follow: a tabletop exercise in which fire officers manage an incident with and without the framework; a transfer study keeping the decision layer and replacing the simulation core with another spreading hazard, the rewriting cost separating architecture from wildfire; and, in the simulator, an ensemble study of decision stability across forecasts and a scale study growing the domain at fixed agent count.

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